

Competitive Programming

Section 75: Master+ Geometry Mixed Contest

Competitive Programming Course

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1 Contest contract

This six-problem contest synthesizes Sections 73–74 with entirely new local statements.

Problem	Main invariant
A	strict hull plus lattice count per edge
B	unimodal triangle area on a convex boundary
C	collision translations form a Minkowski difference
D	weighted active set under inclusive sweep events
E	lengths covered at least once and at least twice
F	strict hull plus logarithmic fan location

2 Suggested use

Allow four hours. Draw all degeneracies before coding: collinear input, duplicate points, end-point contact, equal event coordinates, and zero-area rectangles. For every pointer or event order, write the monotonicity or inclusion statement that makes it legal.

3 How to choose the geometric compression

Statement evidence	First model to test
only extreme points can optimize the objective	strict convex hull
one support point changes as another boundary feature advances	rotating calipers
positions collide after adding a translation	Minkowski difference
objects enter and leave along one coordinate	sweep line and active structure
coverage threshold is “at least one/two”	layered covered-length state
many points ask membership in one convex polygon	fan binary search

The model should reduce the number of candidates or the dimension of active state. If it merely renames the original quadratic search, keep looking.

4 Discovery walkthrough: Robot Collision Translations

Two occupied convex sets collide after translation $\mathbf{t}$ when there exist points $\mathbf{a}$ and $\mathbf{b}$ with $\mathbf{a}+\mathbf{t}=\mathbf{b}$. Rearranging gives

$$\mathbf{t} = \mathbf{b}-\mathbf{a}.$$

Therefore every collision translation lies in $\mathbf{B}-\mathbf{A}$, a Minkowski difference. This algebraic rearrangement is the decisive step; polygon edge merging is only its efficient implementation. Querying a translation becomes ordinary point membership in one precomputed convex polygon.

5 Discovery walkthrough: Double-Painted Map

Ordinary union area stores length covered at least once. Asking for area covered at least twice does not require a separate sweep. Strengthen each node to retain two thresholds: length covered at least once and at least twice. A node's local cover count shifts what its children must provide:

```
cover >= 2: both lengths are the full interval
cover == 1: once is full; twice is children's once-covered length
cover == 0: take both thresholds from the children
```

This is a general state-design habit: when the requested threshold increases, store the small ladder of lower thresholds needed to reconstruct it.

6 Proof map

For hull and Minkowski problems, prove the boundary compression before the pointer or binary search. For sweeps, prove the active-set invariant immediately before each event type. For coverage trees, prove a node formula by cases on its local cover count. For calipers, name the unimodal quantity rather than asserting that a pointer “obviously” moves forward.

7 Stress-oracle map

- A and B: enumerate all points or all vertex triples on tiny inputs.
- C: enumerate every pairwise point difference before taking its hull.
- D: test every horizontal–vertical segment pair directly.
- E: rasterize small integer rectangles and count coverage per unit cell.
- F: compare logarithmic location with a linear edge-sign scan.

Use adversarial collinearity, duplicated raw points, parallel edges, touching boundaries, equal x-events, nested rectangles, and queries on polygon vertices.

8 Submission checklist

- Is polygon orientation known and consistent?
- Are cross products promoted before multiplication?
- Are equal-angle Minkowski edges merged?
- Does event ordering match closed or half-open geometry?
- Do compressed leaves use coordinate differences rather than index counts?
- Are boundary queries classified according to the statement?

9 Exercises

Solve A–F without opening the editorials. The set has exactly six exercises and collectively covers every principal technique from Sections 73–74.